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Checks-in, and Balances: a Matter of Time

Variable Names

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Living registry per Ground Rule 1. Updated as relevant as the model is built. Each variable is a class with the domain shown. Attribute variables may be read as functions of the flight serial n - e.g. $\text{cap}(n)$, $\text{dep}(n)$, $\text{dest}(n)$.

Naming-collision notes. N (total seats) and n (flight serial) are reserved. The count of open check-in desks is therefore written D (not N), and a desk index is written k (not n).

I Schedule, seats & passengers (Steps 1-2)

Variable	Meaning	Class / Domain	Example	First defined
flight	Flight designator (carrier code + number)	String	AI 101	Step 1
n	Flight serial number	Integer, $1 \leq n \leq n_{\text{last}}$	1	Step 1
route	Routing; ordered IATA codes, origin DEL	String DEL-...-XYZ	DEL-CCU-SFO	Step 1
dest	Final destination	String, 3-letter IATA	SFO	Step 1
cap	Capacity (seats offered) of a flight	Integer ≥ 1	288	Step 1
dep	Departure time (SOBT), time of day	HHMM, 24h; $0000 \leq \text{dep} \leq 2359$	0005	Step 1
seat	Seat serial; one per occupiable seat, continuous over all flights in dep order	Integer, $1 \leq \text{seat} \leq N$	1	Step 1
seat_on_flight	Seat ordinal within a flight	Integer, $1 \leq \text{seat_on_flight} \leq \text{cap}$	1	Step 1
N	Total seats in the universe: $N = \sum \text{cap}(n)$	Integer	39610	Step 1
n_last	Number of flights (last serial)	Integer	164	Step 1
LF	Load factor - assumed fraction of seats occupied	Real, $0 < \text{LF} \leq 1$	0.90	Step 2

Variable	Meaning	Class / Domain	Example	First defined
pax	Realised passenger (an occupied seat, exists = 1); per-flight count pax = round(LF x cap)	Integer ≥ 0	259	Step 2
exists	Whether a seat is occupied: exists(seat) in $\{0, 1\}$; a pax is a seat with exists = 1	Binary in $\{0, 1\}$	1	Step 2
entry(pax)	Time a passenger (pax) enters the check-in area; defined for each pax	Timestamp (24h; may fall on day before dep)	20-06-2026 21:30	Step 2
m	Entry-time model label (a distribution over entry(pax))	Categorical/index	M1	Step 2

II Service & queue layer (Steps 3-4)

Variable	Meaning	Class / Domain	Example	First defined
a	Airline (carrier) operating a flight	Categorical (carrier code)	AI	Step 3
t	Clock time within the operating window	Timestamp / minutes	2230	Step 3
D	Open check-in desks for an airline at time t ; D = D(a, t)	Integer ≥ 0 (reserved N)	6	Step 3
k	Check-in desk ("point") index in the open set	Integer, $1 \leq k \leq D$ (reserved n)	1	Step 3
tau (tau)	Service time / processing speed, seconds per pax	Real > 0 (sec.pax)	90	Step 3

Variable	Meaning	Class / Domain	Example	First defined
mu (mu)	Service rate, pax per unit time; mu = $1/\tau$	Real > 0 (pax.time)	40/hr	Step 3
loyaltystatus	Passenger loyalty tier	Integer 0-5: none, silver, gold, platinum, 1K-eq, GS-eq	3	Step 3
class	Cabin class	Integer 1-4: economy, premium economy, business, first	1	Step 3
fastforward	Purchased queue priority (e.g. IndiGo)	Binary in $\{0, 1\}$	0	Step 3
E(p)	Eligible desks a passenger p may use (from class , loyaltystatus , fastforward)	Set subset of open desks of the flight's airline	$\{1, 2\}$	Step 4
w	Wait time endured before service begins	Real ≥ 0 (minutes/seconds)	12 min	Step 4

Author-defined core variables: **flight**, **n**, **route**, **dest**, **cap**, **dep**. *Introduced in Step 1:* **seat**, **seat_on_flight**, **N**, **n_last**. *Introduced for Step 2 (mechanism specified, build pending):* **LF**, **pax**, **exists**, **entry(pax)**, **m**. *Introduced for Steps 3-4 (specification stage):* **a**, **t**, **D**, **k**, **tau**, **mu**, **loyaltystatus**, **class**, **fastforward**, **E(p)**, **w**.